

1

Reflection of Light

In this chapter, we will learn about

- laws of reflection
- images formed by a plane mirror

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1.1 Light rays

The set-up below consists of a black box with 3 LEDs fixed on one side (Fig 1.1a). The opposite side of the box is made of translucent paper. A vertical plate with a hole at the centre is put in the middle of the box. The LEDs form three light spots on the translucent paper. Which pattern in Figure 1.1b does the observer see?

Video

→ This video demonstrates how light travels along a straight line.

Flipped classroom

Watch a video and answer the questions.

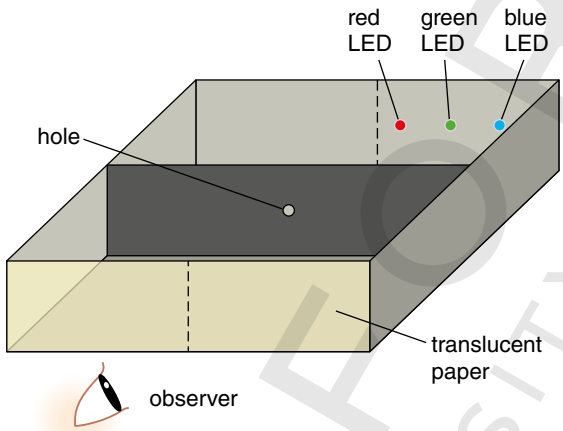


Fig 1.1a *A black box with three LEDs.*

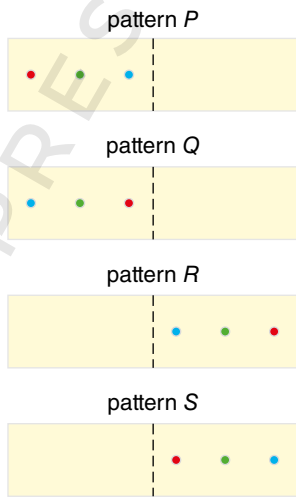


Fig 1.1b *Which pattern does the observer see?*

1 Light rays and light beams

In around 1000 AD, scientist *Alhazen* did a similar experiment. He found that each light spot formed could always be connected to its corresponding light source by a straight line passing through the hole (Fig 1.1c). This provided evidence that light travels along a straight line. This characteristic can also be observed in daily life (Fig 1.1d).

Cross-link

We shall see in Book 3B Ch 6 that this is not strictly correct. ▶

The observer in Fig 1.1a will see pattern *Q*. ▶

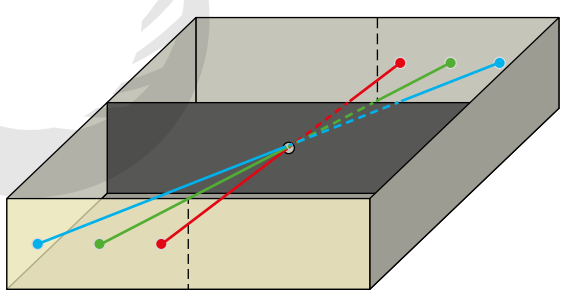


Fig 1.1c *Evidence of light travelling in a straight line.*

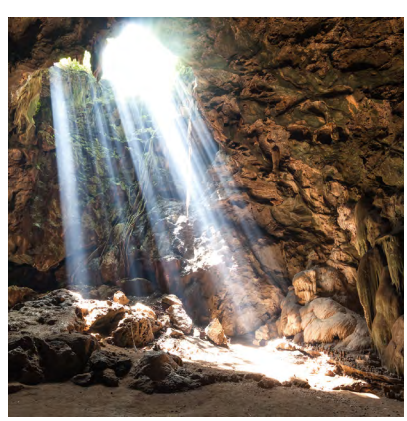


Fig 1.1d *Straight light beams entering a cave.*

Know more

Alhazen (965–1039)

Alhazen was the first to explain that we see things because there is light entering our eyes.

▶ p.33

Alhazen 阿爾哈曾

Based on this characteristic, a ray model was developed to study light. In this model, light consists of **rays** travelling in straight lines. Each ray is represented by a solid straight line with an arrow showing the travelling direction (Fig 1.1e).

A *light beam* is a group of light rays. It may be **parallel**, **divergent** or **convergent**. Figure 1.1f shows how we draw different types of light beams.

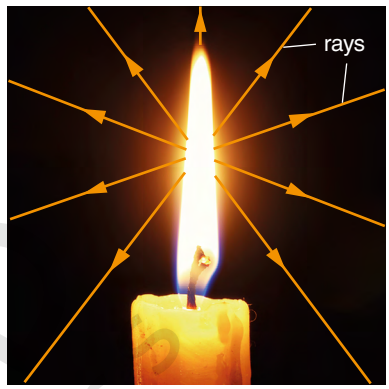


Fig 1.1e *A candle flame emits light rays in all directions.*

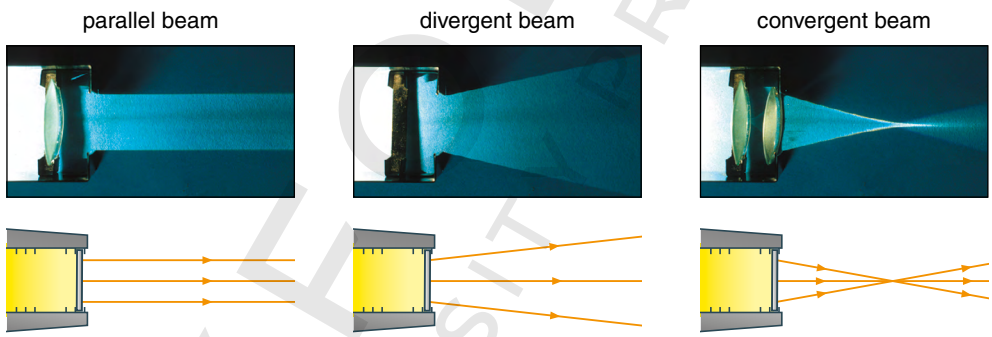


Fig 1.1f *Different types of light beams and their representations by light rays.*

Know more

Some facts about light

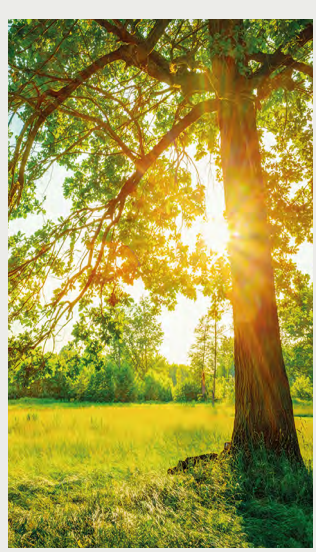
1 Light can travel in a vacuum.

2 Light travels at a very high speed, about 3.00×10^8 m s^{-1} in a vacuum. If you could run as fast as light, you could go around the Earth about 8 times in 1 s!

3 Light carries energy. It can be used in many areas, such as generating electricity and laser cutting. Plants also need light energy to grow.







Know more

Speed of light

People used to believe that light travels at infinite speed. Danish astronomer *Ole Rømer* was the first to think that the speed of light is finite. He made the first quantitative measurements in 1676. The value he found was about 2.2×10^8 m s^{-1}.

ray 光線　light beam 光束　parallel 平行　divergent 發散　convergent 會聚
Ole Rømer 羅默

2 Drawing rays

An object that emits light is a *luminous object*. When the light emitted by an object enters our eyes, we see the object. An object that does not emit light is a *non-luminous object*. We see it if light from a light source bounces off it and enters our eyes. We cannot see it if no light shines on it.

The light from an object may travel in many directions. Only the light entering our eye produces vision. To show graphically how we see a certain point on an object, we draw a **cone of rays** coming from that point to the eye (Fig 1.1g). To show how we see the whole object, we draw two cones of rays from the tips of the object to the eye (Fig 1.1h).

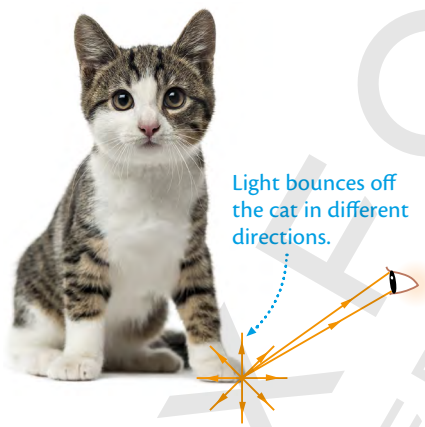


Fig 1.1g *Seeing a point.*

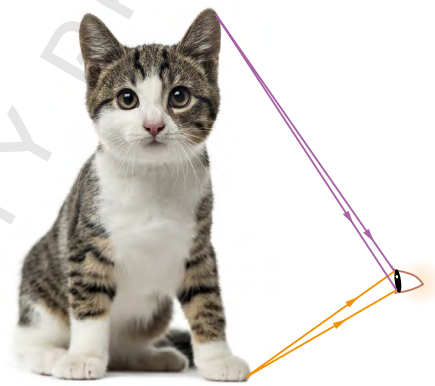


Fig 1.1h *Seeing the whole object.*

The light from a point on a near object reaches our eyes as divergent rays. If the object moves away from us, the rays from the same point will become less divergent. The rays from a point on a distant object (e.g. a point on the Moon) are regarded as parallel (Fig 1.1i).

Emphasize that the parallel rays come from one point on a distant object. Some Ss may wrongly think that the rays come from the whole object, i.e.:

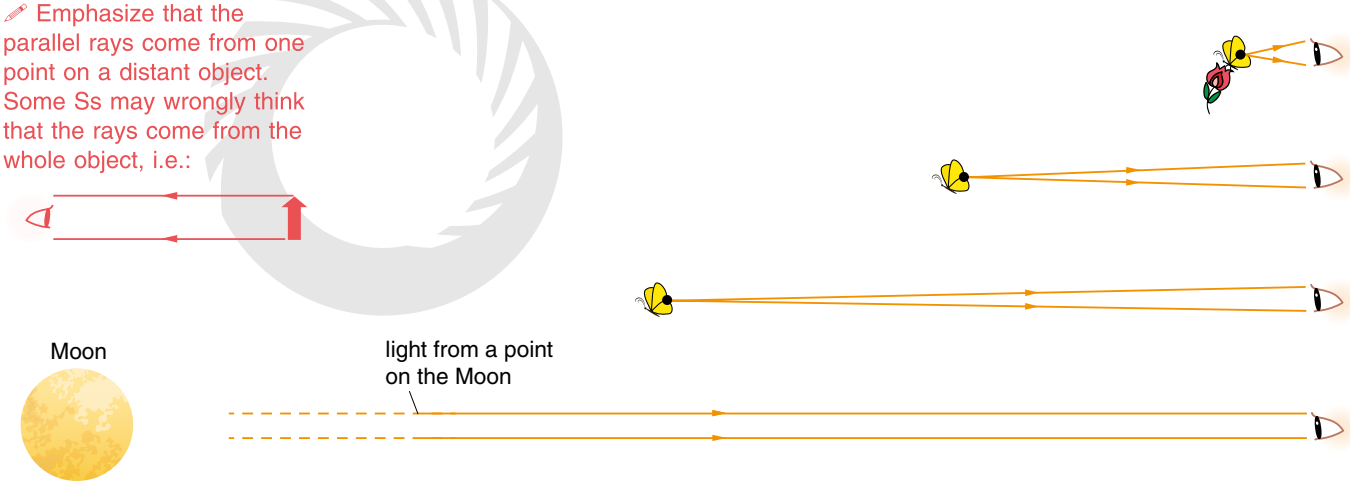


Fig 1.1i *Rays from a point on near and distant objects.*

luminous object 發光體　non-luminous object 不發光體　cone of rays 光錐

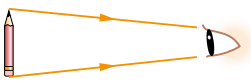
Quick check 1

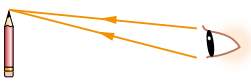
→ E-version in OUP Exercise Platform is also available.

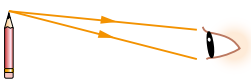
Google Forms | Microsoft Forms

1 *True or false:* It is impossible for us to see non-luminous objects. (T/F)

2 Which of the following diagrams best shows the path of light allowing us to see the tip of the pen?

A 

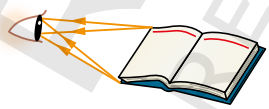
B 

C 

3 Which of the following diagrams best shows the path of light allowing us to see the entire book?

A 

B 

C 

Exercise 1.1

1 

The light from a point on a distant advertising sign reaches our eyes as

A convergent rays.

B divergent rays.

C parallel rays.

D divergent rays at first and parallel rays finally.

2 We can see an apple because

A light from our eyes reaches the apple.

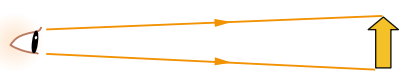
B the apple emits light into our eyes.

C light bounces off the apple into our eyes.

D the apple absorbs light falling on it.

3 Which of the following ray diagrams correctly shows the light rays coming from a point on a distant object to an observer?

A 

B 

C 

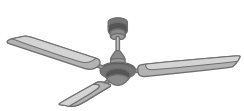
D 

4 In each of the following cases, draw cones of rays to show how the observer sees the whole object.

(a)



(b)





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1.2 Reflection and plane mirror

What a stunning scene of a lake!



Fig 1.2a *The lake surface acts like a plane mirror.*

The water surface reflects light like a plane mirror. Let us learn more about the reflection of light by a plane mirror in this part.

1 Reflection of light

When a ray hits a surface, it may bounce off. This is called **reflection**. Figure 1.2b shows the reflection of light on a **plane mirror** and the corresponding ray diagram. The terms used to describe the reflection of light are given in Table 1.2a.

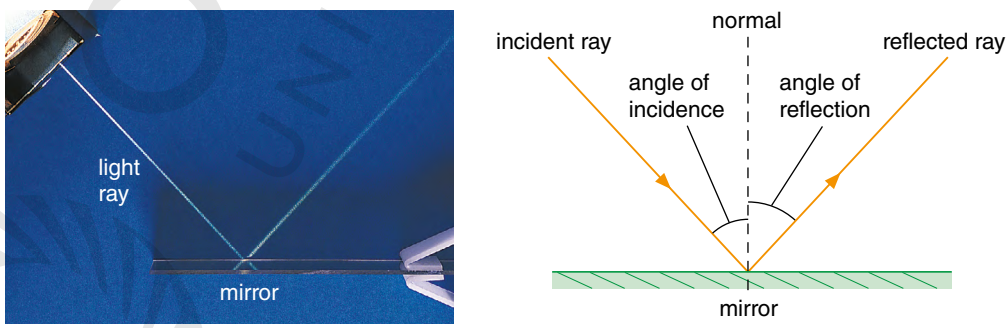


Fig 1.2b *Reflection of light on a plane mirror.*

Incident ray	The light ray that approaches the mirror
Reflected ray	The light ray that leaves the mirror after reflection
Normal	An imaginary line perpendicular to the plane mirror at the point where the incident ray hits the mirror, represented by a dotted line
Angle of incidence	The angle between the incident ray and the normal
Angle of reflection	The angle between the reflected ray and the normal

Table 1.2a *Describing reflection of light.*

Caution

Do **not** use the angle between the ray and the mirror as the angle of incidence or angle of reflection.

reflection 反射　plane mirror 平面鏡　incident ray 入射線　reflected ray 反射線
normal 法線　angle of incidence 入射角　angle of reflection 反射角

DSE goal
State the laws of reflection.

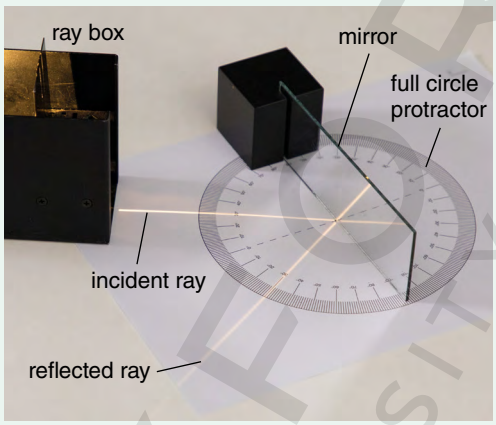
2 The laws of reflection

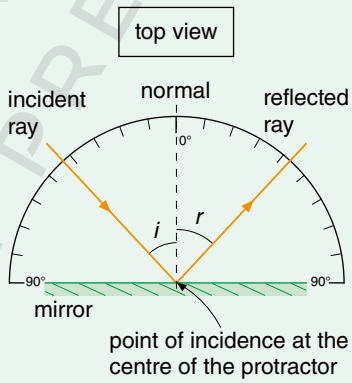
Experiment 1a / Reflection of light

1 Put a plane mirror on the 90°–90° line of a full circle protractor.

2 Direct a light ray at the centre of the protractor. Measure the angle of incidence i and the angle of reflection r.

3 Repeat step 2 by changing i.





Discussion

How is r related to i?

Simulation Video

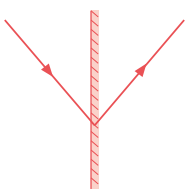
→ This simulation is a 'virtual experiment' of the reflection of light.

→ This video demonstrates Expt 1a.

Do not touch the hot ray-box lamp.

A full circle protractor can be found in Practical Workbook 3A.

Some Ss may draw a ray diagram like:

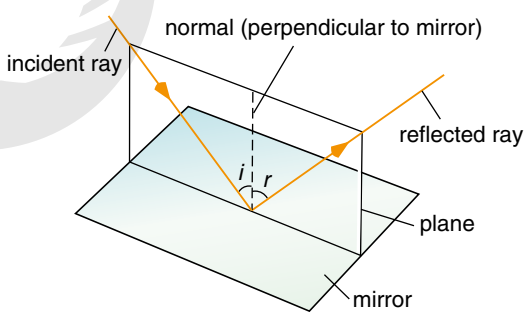


They misunderstand that they see the 'reflected ray' when looking into the mirror. Ts may need to further explain the meaning of reflected ray or use mirror strip of smaller heights in Expt 1a.

In the above experiment, we find that r is always the same as i. This agrees with the **laws of reflection** (Fig 1.2c):

- **The incident ray, the reflected ray and the normal all lie in the same plane.**
- **The angle of reflection *r* is always equal to the angle of incidence *i*.**

(i)



(ii)

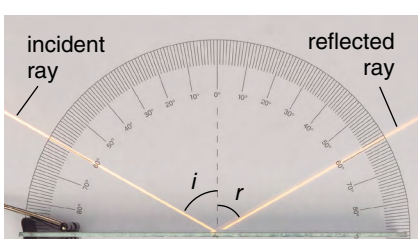


Fig 1.2c *The laws of reflection.*

laws of reflection 反射定律

Example 1 | Applying the laws of reflection

A ray of light hits a plane mirror (Fig a). The angle between the incident ray and the mirror is 22°. Find the angle of reflection r and complete the ray diagram.

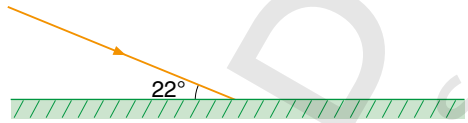


Fig a

Solution

i is the angle between the incident ray and the normal (Fig b).

$i = 90° - 22° = 68°$

By the laws of reflection, $r = i = 68°$

Figure c shows the complete ray diagram.

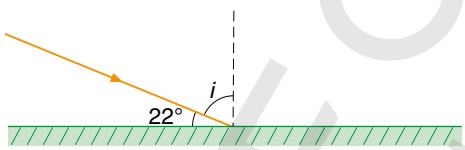


Fig b

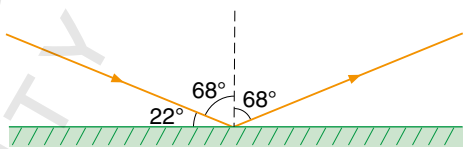
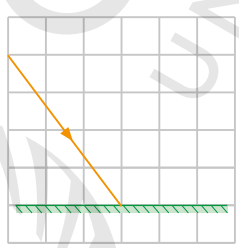


Fig c

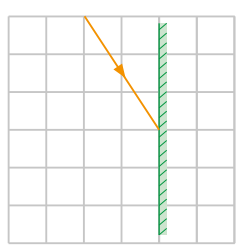
Example 2 | Drawing the reflected ray

In each of the cases below, a ray of light is directed to a plane mirror. Draw the normal and the reflected ray in each case.

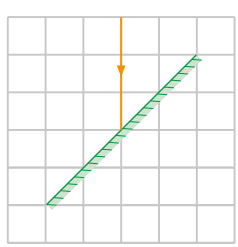
(a)



(b)

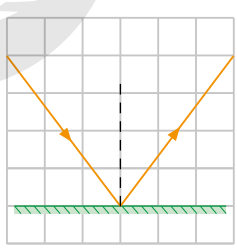


(c)

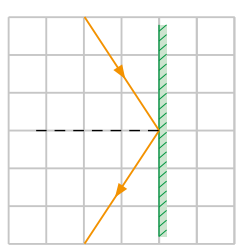


Solution

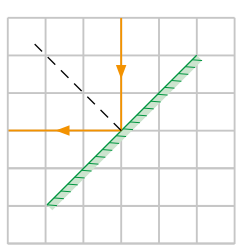
(a)



(b)



(c)



Light follows the same path if the direction of light is reversed. If we direct a light ray along the original reflected ray, the new reflected ray will travel along the original incident ray. This is called the **reversibility of light** (Fig 1.2d).

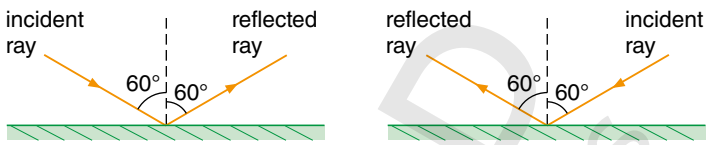


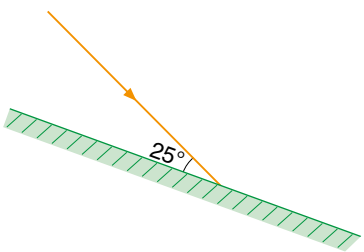
Fig 1.2d *The reversibility of light.*

Quick check 2

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

(**For Q1–2.**) A ray of light hits a plane mirror as shown. The angle between the incident ray and the mirror is 25°.



1 What is the angle of incidence?

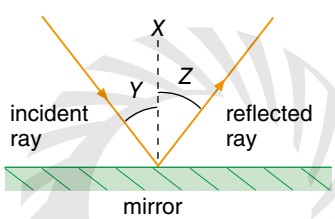
A 25°
B 65°
C 75°
D 155°

2 What is the angle of reflection?

A 25°
B 65°
C 75°
D 155°

Learn by practice 1

1 A light ray is reflected by a plane mirror as shown.

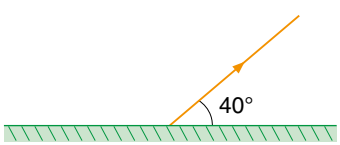


(a) Name line *X*, angles *Y* and *Z*.

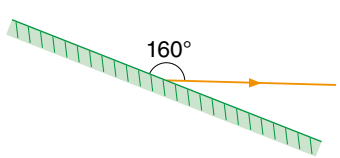
(b) What is the relationship between *Y* and *Z*?

2 Draw the normal and the incident ray in each of the following diagrams.

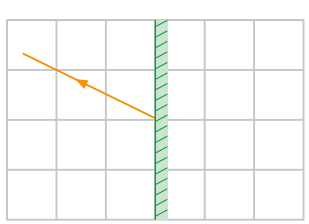
(a)



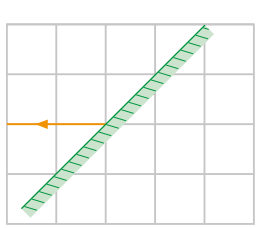
(b)



(c)



(d)



DSE goal

Construct images formed by plane mirror graphically.

DSE exam

18(1A)Q20, 20(1A)Q17, 22(1A)Q20

3 Image formed by plane mirror

With some transparent plates and a smartphone, you can make an animation that floats in the air (Fig 1.2e)! How is this possible? Before answering this question, we shall learn how an **image** is formed by a plane mirror.

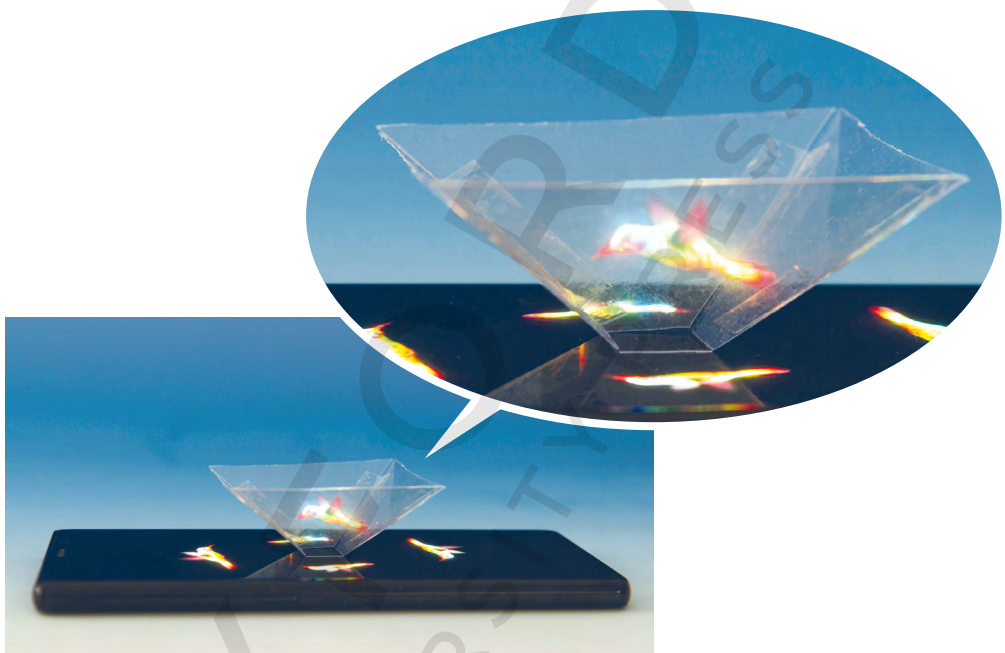


Fig 1.2e *Animation floating on a mobile phone.*

a How a plane mirror forms an image

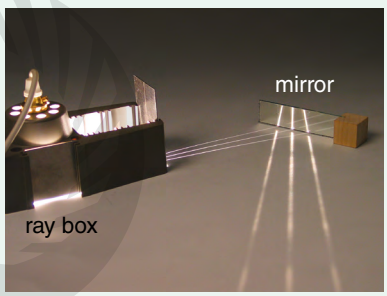
Video

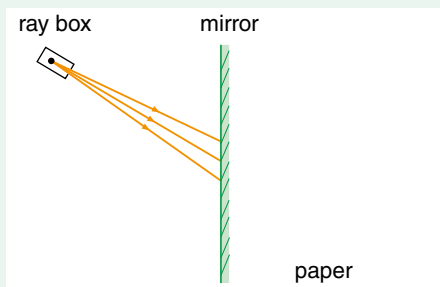
→ This video demonstrates Expt 1b.

Do not touch the hot ray-box lamp.

Experiment 1b An image formed by a plane mirror

1 Direct a set of divergent rays at a plane mirror on a piece of paper.





2 Use solid lines to trace the incident and the reflected rays on the paper.

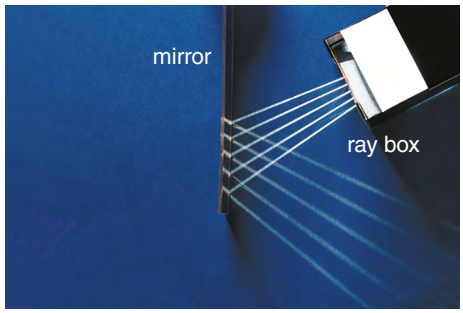
3 Use dotted lines to trace the reflected rays backwards.

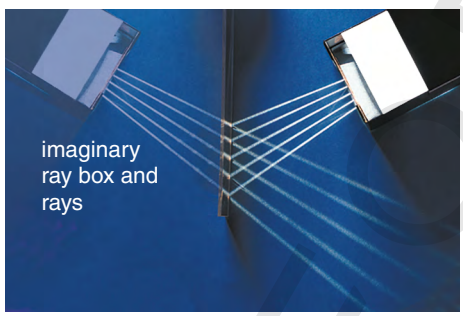
Discussion

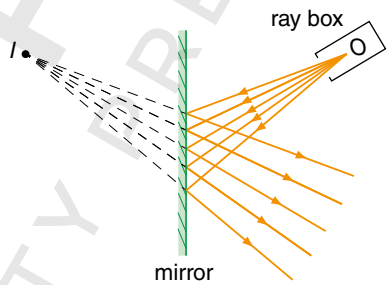
What do you notice about the reflected rays when they are traced backwards?

image 像

In Experiment 1b, when divergent rays are reflected by the mirror, the reflected rays appear to come from a point behind the mirror (Fig 1.2f).



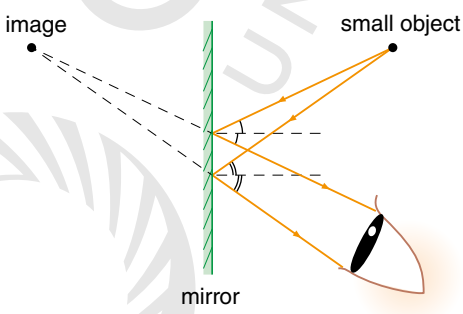


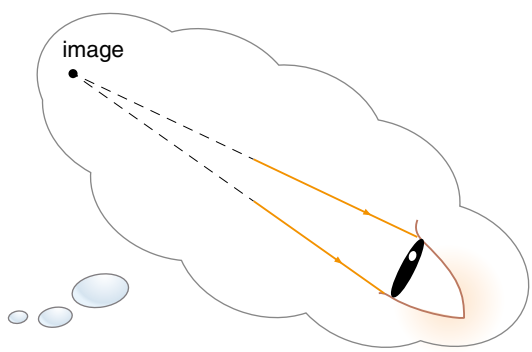


The imaginary ray box and rays aim to illustrate that the reflected rays appear to come from a point behind the mirror. They cannot actually be seen from the angle that this photo was taken.

Fig 1.2f *Divergent rays reflected by a mirror appear to come from a point behind the mirror.*

If we look at a small object in a mirror, light rays from the object are reflected by the mirror and enters our eyes (Fig 1.2g). Like Figure 1.2f, the reflected rays appear to come from a point behind the mirror. We perceive the reflected rays as coming straight from this point, i.e. the position of the object's image.





✎ It doesn't matter how the light rays travel before entering our eyes. Light rays can travel along different paths before entering our eyes, but we perceive them as coming from the same point.

Fig 1.2g *We perceive the light rays as coming from the image behind the mirror.*

Notice that:

1 No rays actually come from the image. This kind of image is called a **virtual image**.

2 The rays behind the mirror do not actually exist. They are represented by **dotted lines without arrows** in ray diagrams.

virtual image 虚像

→ This video demonstrates Expt 1c.

Video

b Properties of an image

Experiment 1c Location and size of an image formed by a plane mirror

1 Place a transparent plate vertically along a line on grid paper.

2 Place a small object in the front region and observe its image (Fig a). Locate its image formed by the plate by putting an identical object at the back region to overlap with the image.

3 Repeat step 2 by replacing the two small objects with two D cells. Compare the sizes of the image and the cell in the back region.

4 Place a pencil in the front region (Fig b). Use two markers to locate the two ends of the image. Measure the lengths of the pencil and its image.

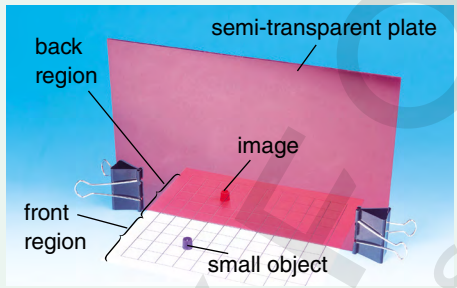


Fig a

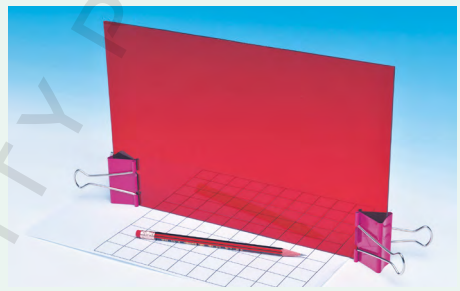


Fig b

Discussion

1 Is the line joining the object and the image perpendicular to the plate?

2 What can you conclude about the position and the size of the image?

From Experiment 1c, we conclude that:

1 the line joining the image *I* and the object *O* is perpendicular to the mirror *MN*, i.e. $OI \perp MN$ (Fig 1.2h);

2 the image is as far behind the mirror as the object in front, i.e. image distance is equal to object distance (Fig 1.2h);

3 the image is of the same size as the object (Fig 1.2i).

Note that a virtual image is represented by a **dotted arrow**.

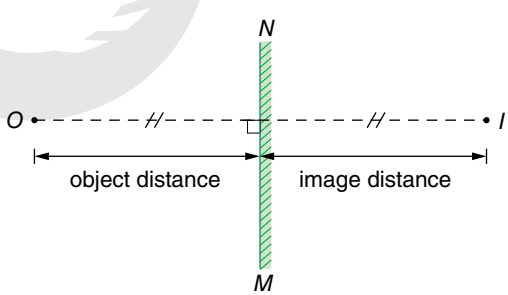


Fig 1.2h Location of the image.

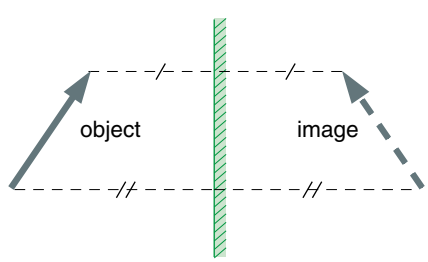


Fig 1.2i Size of the image.

Know more

Transparent plate

When light falls on a transparent plate, part of it passes through the plate and part of it is reflected.

▸ p.33

The transparent plate used in this experiment are actually acrylic plates. They are available from some advertising production companies.

After Ss learn the rules in locating the image, ask them to change the orientation of the plate, e.g. along the diagonal of the grids. Some Ss may not use the rules correctly in this case.

The distances are measured from the front of the mirror in a ray diagram. ▸

Know more

Location of an image and laws of reflection

The location of the image formed by a plane mirror can be deduced from the laws of reflection and some mathematics.

▸ p.34

Know more

Words on an ambulance

The word ' 救護車 ' (ambulance in Chinese) on an ambulance is laterally inverted, so that it will be read easily in the rear-view mirror of the vehicle in front of the ambulance.



We can observe one more property of the image in daily life. Kick your left leg in front of a plane mirror and your image will appear to kick its right leg (Fig 1.2j). The image is **laterally inverted**.



Fig 1.2j The image formed by a mirror is laterally inverted.

The properties of the image formed by a plane mirror:

- **virtual**
- **behind the mirror**
- **image distance is equal to object distance**
- **the same size as the object**
- **laterally inverted**

Example 3 Size and orientation of an image

A girl 1.4 m tall is 1 m in front of a mirror. She wears a T-shirt as shown.



(a) What is the distance between the girl's image and the mirror?

(b) What is the height of the girl's image?

(c) What will happen to the size of the girl's image if the girl walks towards the mirror? Explain your answer briefly.

(d) Sketch the image of the T-shirt as seen by the girl.

Solution

(a) Image distance = object distance = 1 m

(b) Height of image = height of girl = 1.4 m

(c) The size of the image will be unchanged since it depends only on the size of the girl, not her distance from the mirror.

(d)



laterally inverted 橫向倒置

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) When an object is placed in front of a plane mirror, the image formed has the same size as the object. (T/F)

(b) The image formed by a plane mirror is on the mirror surface. (T/F)

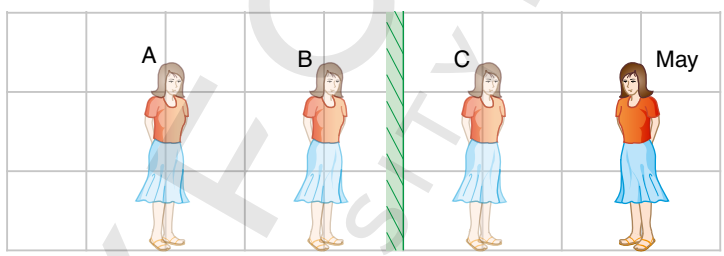
(c) No light ray actually comes from the image formed by a plane mirror. (T/F)

2 A card with letters 'Phy' is put in front of a plane mirror. Which of the following best shows the image of the letters?

A Phy B ɥdɟ C Phy

3 May is 3 m from a mirror.

(a) Where is May's image?



(b) Peter is 1 m away from May. What is the distance between the image of Peter and the image of May?

Learn by Practice 2

1 Two light rays leave object *O* as shown (Fig a).

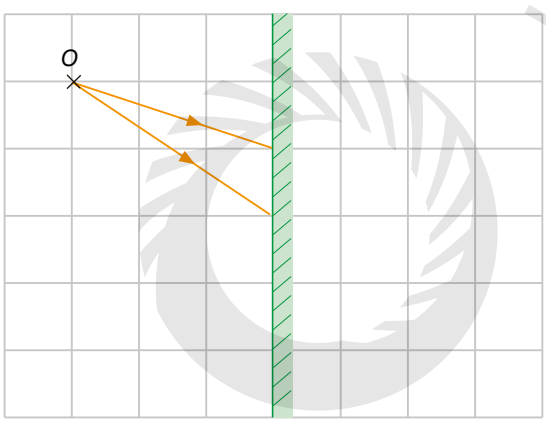


Fig a

(a) Draw the reflected rays.

(b) Mark the position of the image *I* of object *O*.

2 A card with a letter 'B' printed on it is placed in front of a plane mirror (Fig b). Sketch the image of the letter.

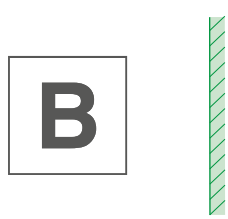


Fig b

3 A boy 1.5 m tall stands 1.8 m in front of a plane mirror.

(a) How tall is his image?

(b) How far is the boy from his image?

(c) How will the answers to (a) and (b) change if he walks 1 m away from the mirror?

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→ Teaching notes and exercises are available on OUP web.

C Drawing an image

We can draw and locate an image by using its properties.

Skill *Drawing*

Drawing an image formed by a mirror

1 Draw a dotted line from a tip of the object to pass through the mirror perpendicularly.

2 Apply image distance = object distance to locate the image of the tip.

3 Locate the image of the other tip of the object. Join the two image points with a dotted arrow to draw the image.

Tip

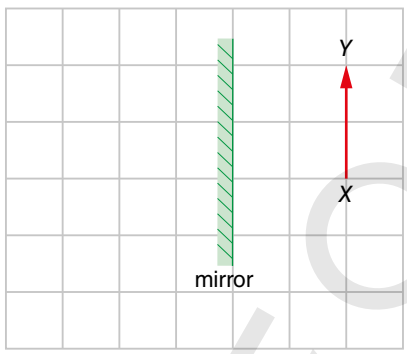
When the mirror is inclined at 45°, the image of a horizontal object is vertical, and vice versa.

Example 4 | Location of images

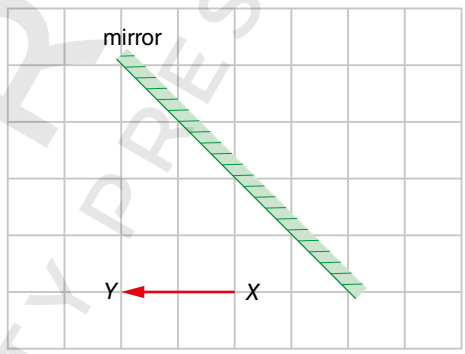
Repeat Experiment 1c on p.12 by putting the plate along the diagonal of the grids.

An object XY is placed in front of a plane mirror. In the following cases, locate the images of points X and Y. Hence draw the image of XY.

(a)

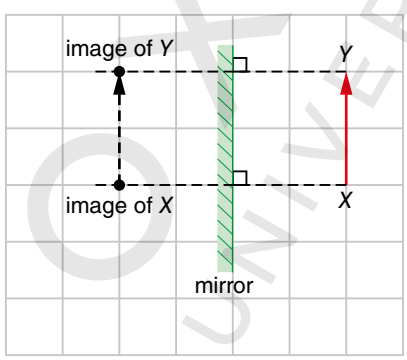


(b)

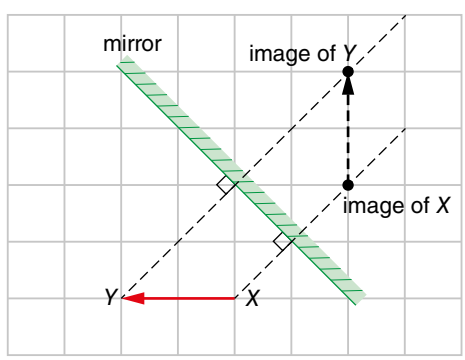


Solution

(a)



(b)



Sometimes, an object may not be right in front of a mirror. We can extent the mirror with a dotted line to help find the image (Fig 1.2k).

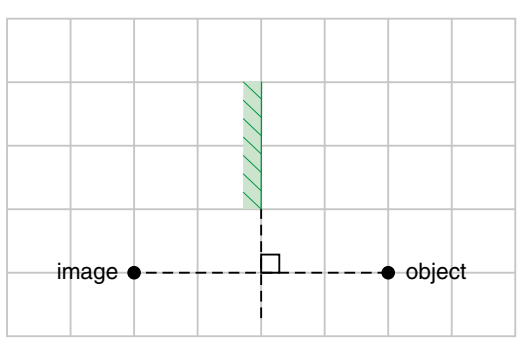


Fig 1.2k *Locating the image of an object not right in front of a mirror.*

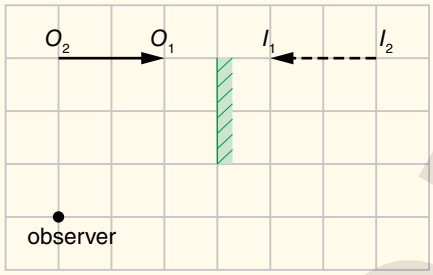
d Seeing an image

Follow the steps below to show if an observer can see an image in a mirror.

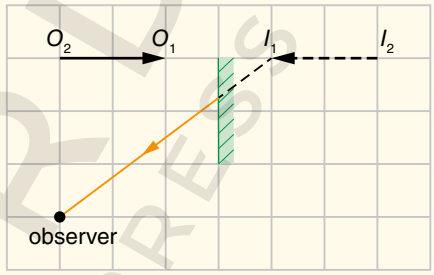
→ Teaching notes and exercises are available on OUP web.

Skill
Drawing

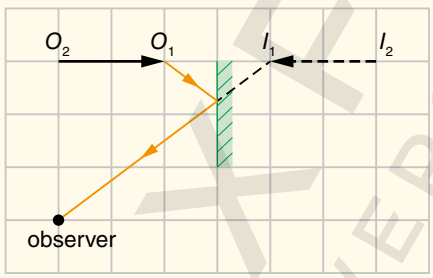
See an image or not



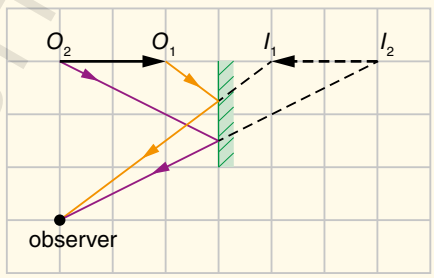
① Locate the image I_1I_2 of the object O_1O_2 by making use of the properties of the image.



② Draw a ray from I_1 to the observer. Use a solid arrowed line to indicate a real light ray. Use a dotted line without an arrow to indicate a virtual ray.



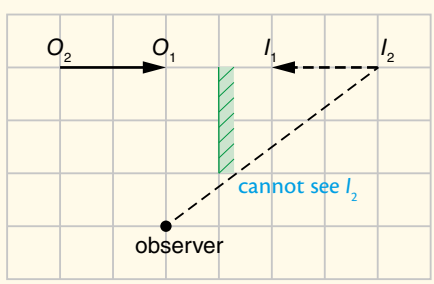
③ Draw a ray from O_1 to the point where the ray from I_1 meet the mirror. This shows that the observer can see I_1.



④ Repeat steps ② and ③ for I_2. This shows that the observer can see the whole image I_1I_2.

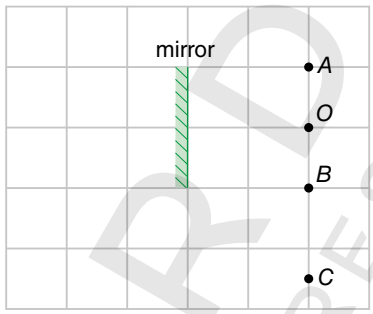
Note the following:

1 Solid lines are used for objects and real light rays.
2 Dotted lines are used for virtual images and virtual rays.
3 Real rays have arrows while virtual rays do not.
4 For simplicity, a single ray instead of a cone of rays is drawn from each point to the observer.
5 If the line joining an image point, say I_2, and the observer does not pass through the mirror, the light from O_2 cannot be reflected by the mirror to the observer, so the observer cannot see I_2.

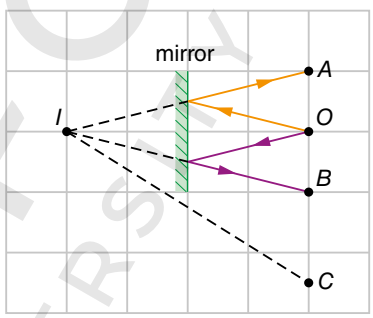


Example 5 | Who can see the image?

Point object O is put in front of a plane mirror. Can students A, B and C see the image of O?



Solution



∴ A and B can see the image while C cannot.

Tip
The position of I is independent of the position of the observer.

Think about...

In which direction should C move in order to see I?

Know more

Video

→ This video shows a coin box which makes use of reflection of light.

Applications of plane mirror

Plane mirrors are widely used in daily life. Below are some examples.



(i) Dentist mirror.



(ii) Wall mirrors.



(iii) Car rear-view mirror.

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Example 6 Position of a dressing mirror

A girl 78 cm tall stands in front of a vertical mirror (Fig a). Her eyes are 8 cm below the top of her head. She can just see her feet at the bottom edge of the mirror when she is 60 cm from it (Fig b).



Fig a

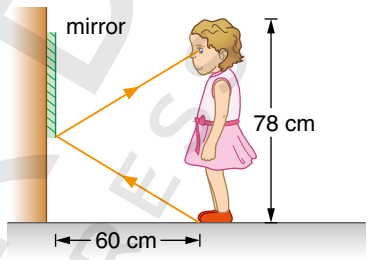


Fig b

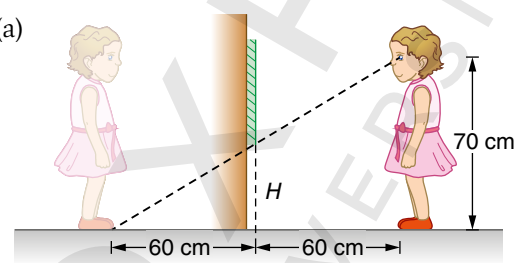
(a) How high above the floor is the bottom edge of the mirror?

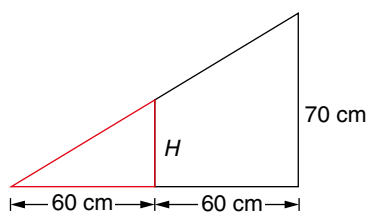
(b) What is the minimum length of the mirror if the girl can just see the image of her whole body in the mirror?

Solution

(a)

The girl's eyes are 70 cm (= 78 – 8) above the floor. ▶





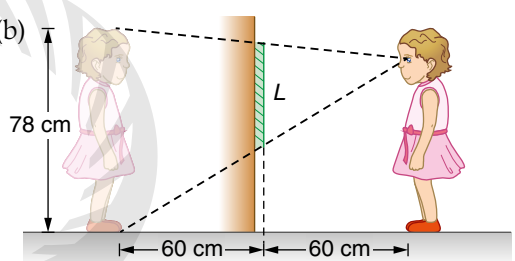
By similar triangles,

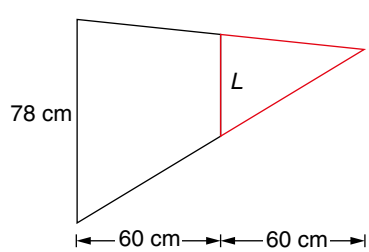
$$\frac{H}{70} = \frac{60}{60 + 60} = \frac{1}{2}$$

$$H = 35 \text{ cm}$$

The bottom edge of the mirror is 35 cm above the floor.

(b)





By similar triangles,

$$\frac{L}{78} = \frac{60}{60 + 60} = \frac{1}{2}$$

$$L = 39 \text{ cm}$$

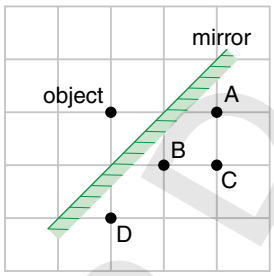
The minimum length of the mirror is 39 cm.

Google Forms Microsoft Forms

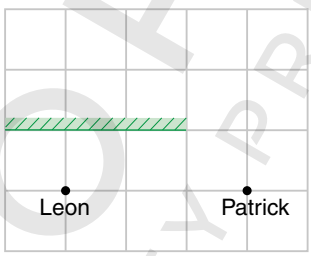
Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Where is the image of the object?



2 Leon and Patrick look in a plane mirror placed directly in front of Leon.

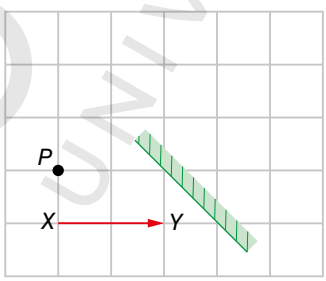


(a) Can Leon see (i) himself; (ii) Patrick in the mirror?

(b) Can Patrick see (i) himself; (ii) Leon in the mirror?

Learn by Practice 3

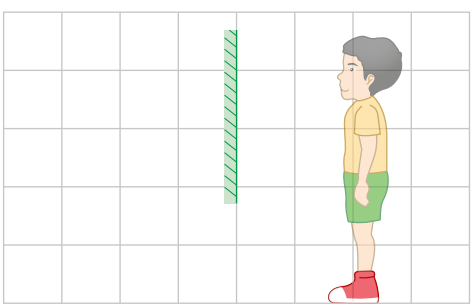
1 In each of the following cases, object *XY* is placed in front of a plane mirror.



(a) Draw the image of *XY* formed by the mirror.

(b) Can an observer at point *P* see the images of points *X* and *Y*? Draw light rays to explain your answer.

2 Danny stands in front of a plane mirror. He is 1.78 m tall. He can see the image of his whole body in the mirror.



(a) Draw light rays to show how he sees the image of his shoes.

(b) His eyes are 1.7 m above the floor.

(i) What is the maximum height of the bottom edge of the mirror above the floor?

(ii) What is the minimum height of the top edge of the mirror above the floor?

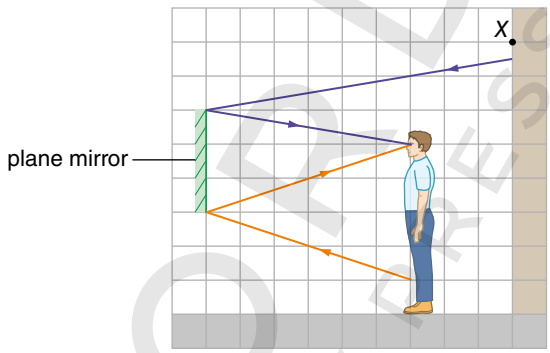
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Example 7 Walking in front of a plane mirror

A man stands in front of a vertical plane mirror. He cannot see his shoes nor point X in the mirror. If he walks towards the mirror,

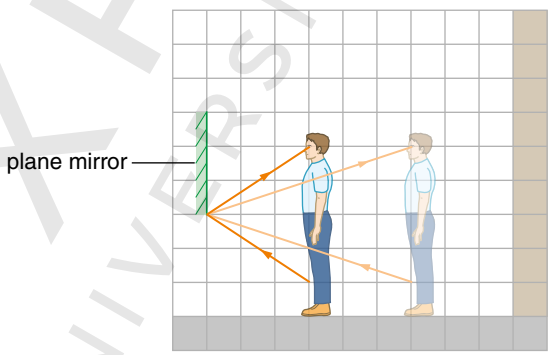
(a) can he see his shoes in the mirror?

(b) can he see point X in the mirror?



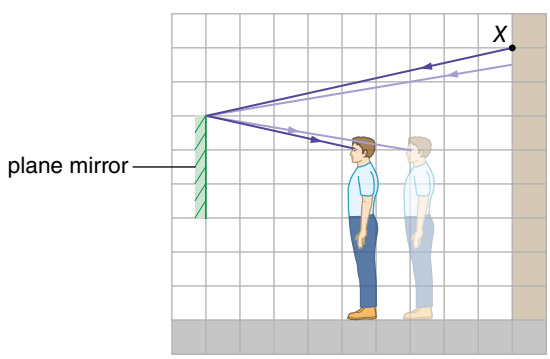
Solution

(a)



When he walks towards the mirror, he can only see the same point on his body at the bottom edge of the mirror, so he cannot see his shoes.

(b)



When he walks towards the mirror beyond a certain point, he can see point X in the mirror.

▶ Exercise 1.2 Q8 (p.25)

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Video

→ The video 'Magic trick using reflection of light' shows a demonstration of a magic trick using reflection of light.

→ The video 'Animation projector' shows how the animation projector is made.

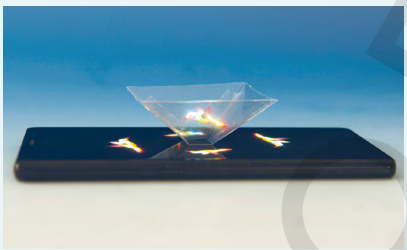
→ The video 'Candle flame in water' demonstrates how the illusion of a burning candle in water is produced.

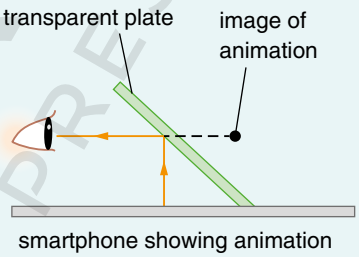
Applications of transparent plates

A transparent plate forms images like a plane mirror but allows some light to pass through it at the same time (see p.33 for detail). It has many applications.

Animation projector

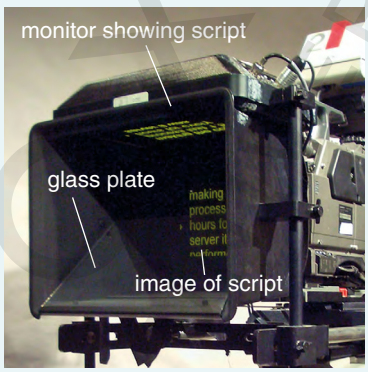
When a transparent plate is put at 45° on a smartphone showing animation, the image of the animation will be formed in mid-air. By putting 4 such plates on the phone, the animation can be seen from different directions.

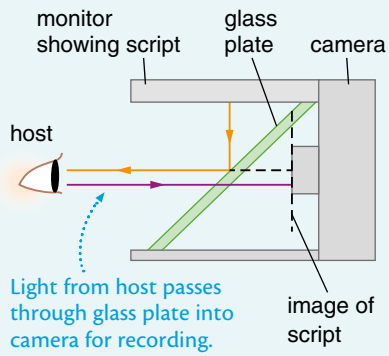




Teleprompter

A *teleprompter* is installed in front of a camera. The transparent glass plate reflects the script to the host. This allows the host to read the script and look directly into the camera at the same time.

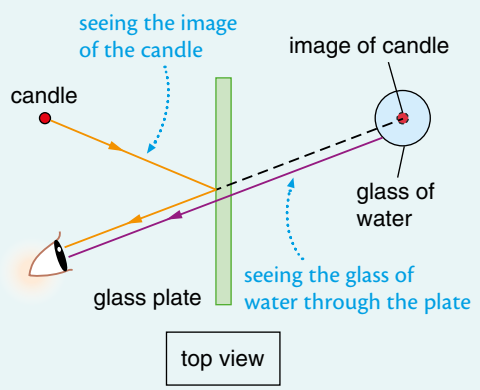




Illusion

We can produce the illusion of a burning candle in water using the following set-up.





teleprompter 提詞機

☆ HKDSE 2020 Paper 1A Q17

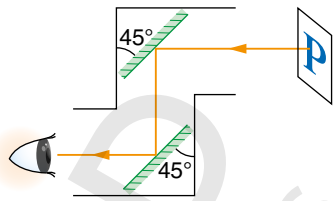
→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Periscope

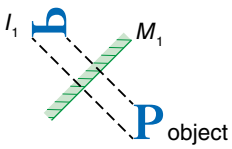
Kate uses a periscope to observe a letter P. Which image does she see?

A B C D



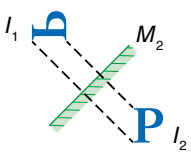
Solution

First, sketch the image I_1 formed by the upper mirror M_1.



Using a 2D diagram is easier than considering the 3D situation. Since we only want to know the orientation of the image, we draw the object at a position closer to the mirror.

The image I_1 is the object for the lower mirror M_2.
The image I_2 formed by M_2 is the answer.



∴ The answer is C.

Revision 1 Q14 (p.30)

We can see that, if there are two mirrors facing each other, the image formed by one mirror may become the object for the other mirror.

4 Regular and diffuse reflection

When a parallel beam of light falls on a **flat and smooth surface** (e.g. a plane mirror), the reflected beam is also parallel (Fig 1.2l). This is called **regular reflection**.

When a parallel beam of light falls on a **rough surface**, it is reflected in different directions (Fig 1.2m). No reflected beam can be seen. This is called **diffuse reflection**.

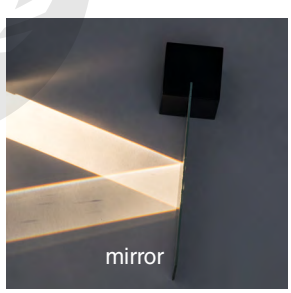


Fig 1.2l Regular reflection occurs on a smooth surface (e.g. a mirror).

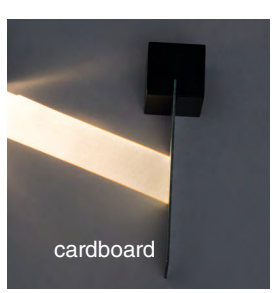


Fig 1.2m Diffuse reflection occurs on a rough surface (e.g. a cardboard).

regular reflection 單向反射　　diffuse reflection 漫反射

Table 1.2b compares regular reflection and diffuse reflection.

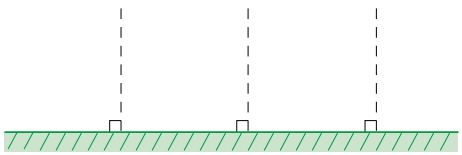
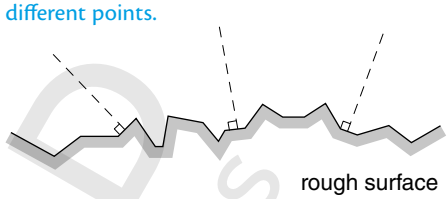
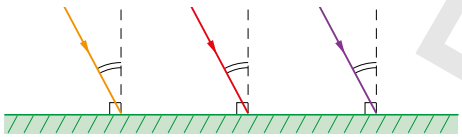
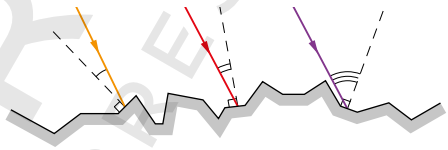
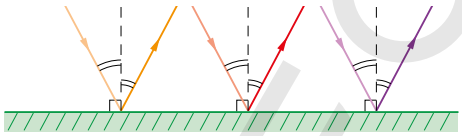
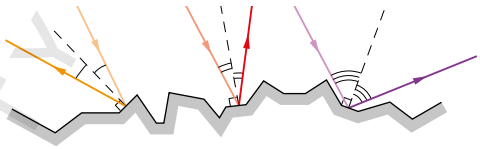
Regular reflection	Diffuse reflection
① Normals are all parallel at different points.  	① Normals point in different directions at different points. 
② Parallel rays fall on the surface. 	② Parallel rays fall on the surface. 
③ The reflected rays are parallel. 	③ The rays are reflected in different directions. 

Table 1.2b *Regular reflection and diffuse reflection.*

Note that:

1. The laws of reflection apply to both regular and diffuse reflection.
2. Regular reflection enables a clear image to be formed while diffuse reflection does not (Fig 1.2n).

To take a photo like (i), you may position your camera just above a puddle on the ground. When the ground becomes dry, take a photo at the same position again to get a photo like (ii).

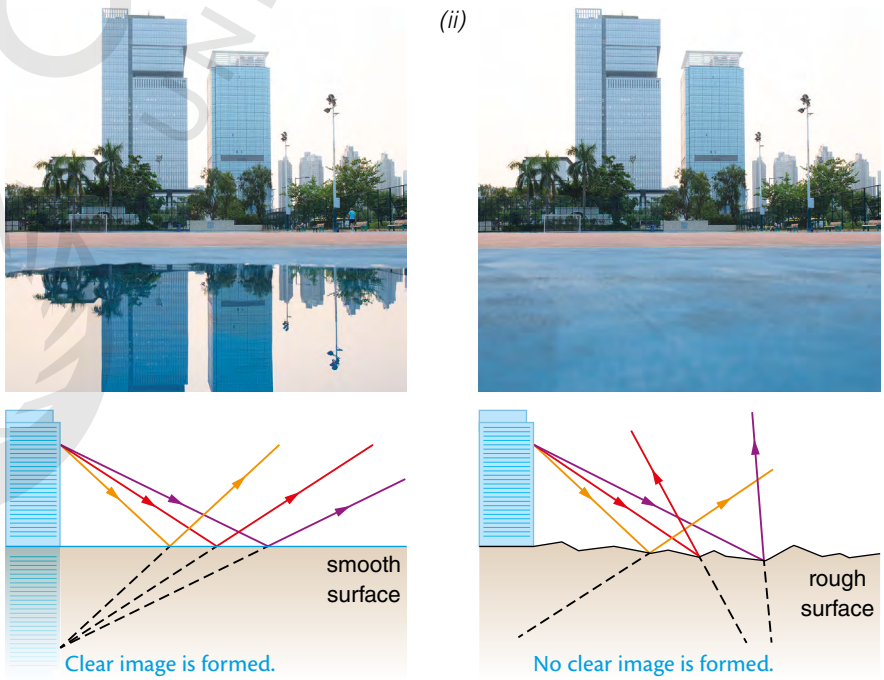


Fig 1.2n *(i) Regular reflection and (ii) diffuse reflection.*

Know more

Glare

Sometimes, *glare* on the blackboard makes it difficult to read the words written on it. Do you know how glare is formed?



▸ p.34

Google Forms

Microsoft Forms

Most surfaces (e.g. paper) are not as smooth as we think (Fig 1.2o). Diffuse reflection occurs on these surfaces, so we can see them from all directions.

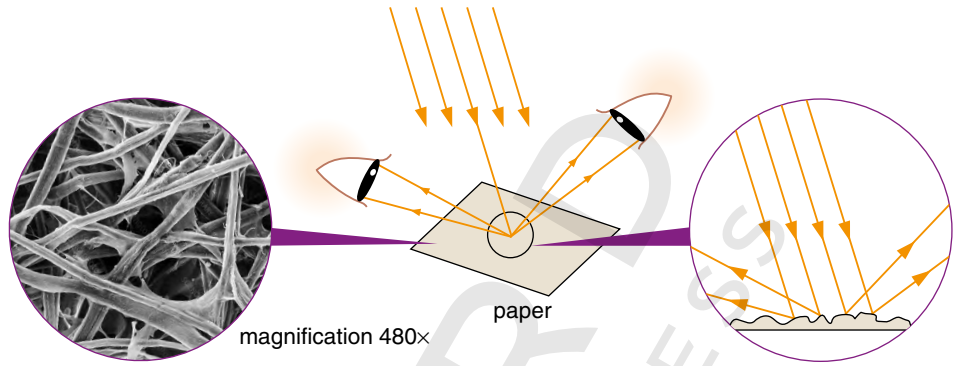


Fig 1.2o *Diffuse reflection allows us to see the paper from different directions.*

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) Regular reflection occurs on a flat and smooth surface. (T/F)

(b) No clear image is formed in diffuse reflection. (T/F)

(c) Diffuse reflection does not obey the laws of reflection. (T/F)

(d) A plain paper does not produce a clear image because it does not reflect light. (T/F)

Exercise 1.2

1 The diagram shows the image of a digital clock formed by a plane mirror (Fig a). Which of the following shows the actual time at this instant?



Fig a

A 

B 

C 

D 

2 Which of the following best explains why a clear image cannot be produced by reflection on a piece of paper?

A Normals point in different directions at different points on the paper.

B The light beams falling on the paper are not parallel.

C Reflection on the paper does not obey the laws of reflection.

D The paper does not reflect light.

3 A ray is reflected by a plane mirror (Fig b). What is the angle of reflection?

A 30°

B 45°

C 60°

D 90°

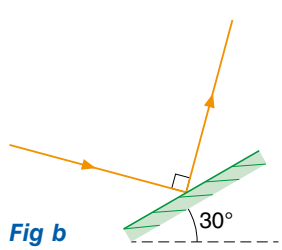


Fig b

glare 眩光

*4 A ray strikes a plane mirror (Fig c). If the mirror is turned clockwise by 20°, what is the angle of reflection?

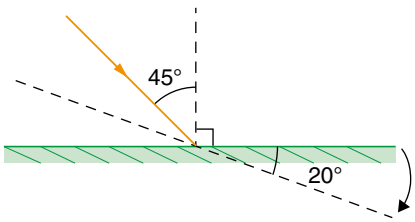


Fig c

A 20° B 25°
C 60° D 65°

*5 A girl looks in a tilted plane mirror (Fig d). Where is the image of her nose tip formed by the mirror?

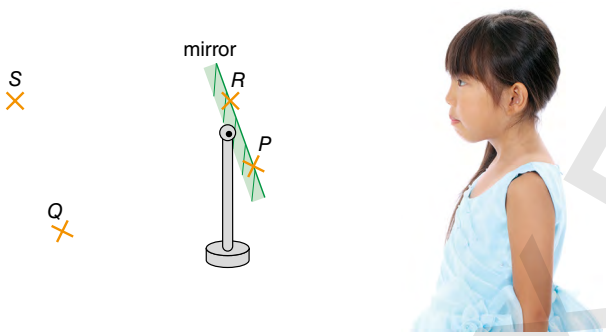
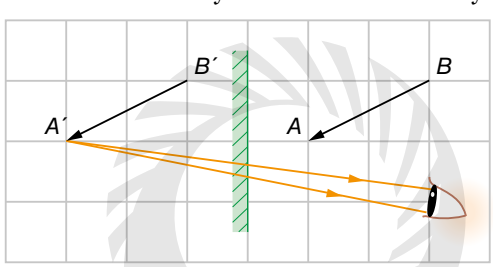


Fig d

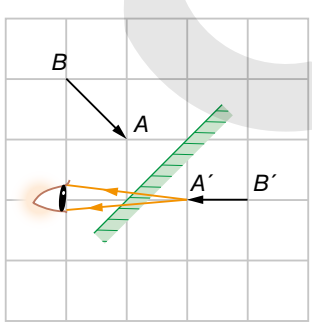
A *P* B *Q*
C *R* D *S*

*6 The following ray diagrams are drawn incorrectly. Redraw them to show the correct image of *AB* and how a cone of ray from *A* enters the eye.

(a)



(b)



*7 Harry is 1 m in front of a plane mirror and he sees a monster in the mirror (Fig e). The height of the image of the monster is 0.8 m. Calmly, he looks behind and estimates that the monster is 2 m away from him.



Fig e

(a) State two properties of the image formed by a plane mirror.

(b) What is the height of the monster?

(c) What is the distance between Harry and the image of the monster?

(d) How will the image of the monster change if Harry walks towards the mirror?

*8 Cally faces a plane mirror (Fig f). *XY* is a wall behind her.

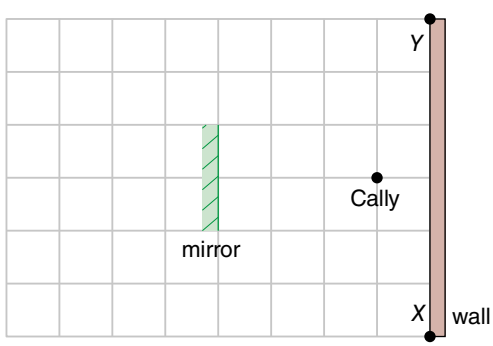


Fig f

(a) Highlight the portion of the wall that she can see in the mirror.

(b) Should she walk towards or away from the mirror in order to see the image of point *Y*? Draw light rays to explain your answer.

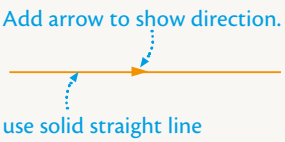
(c) Besides the method in (b), suggest another method that allows her to see the image of point *Y*.

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Key concepts 1

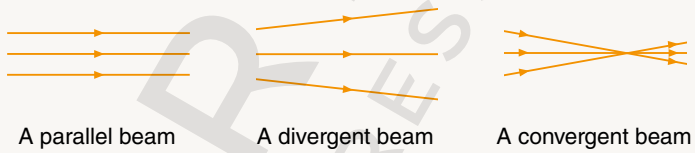
1 Light rays

Light is represented by rays. A ray is a straight line with an arrow.

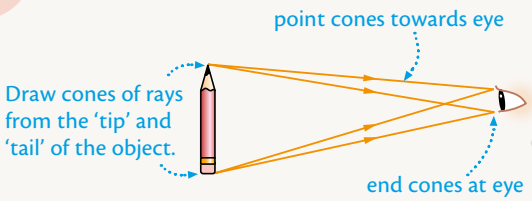


2 Light beams

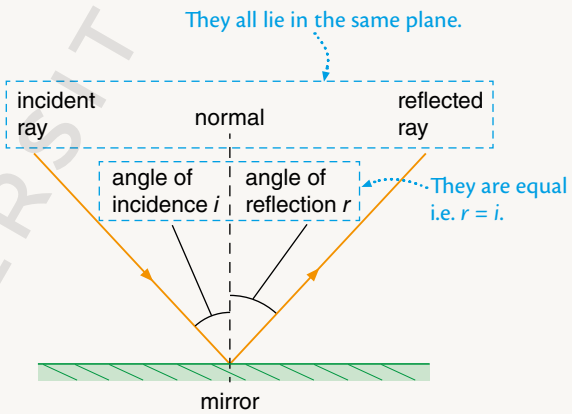
A light beam is represented by a number of rays.



3 How we see things



4 The laws of reflection



5 Images formed by a plane mirror

The properties of the image:

laterally inverted

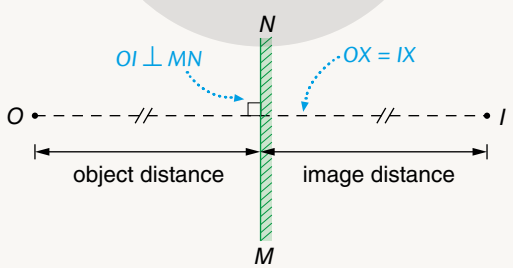
behind the mirror

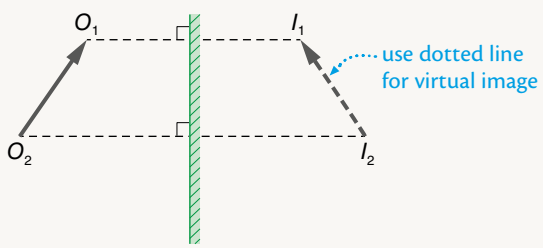
virtual

image distance is equal to object distance

of the same size as the object

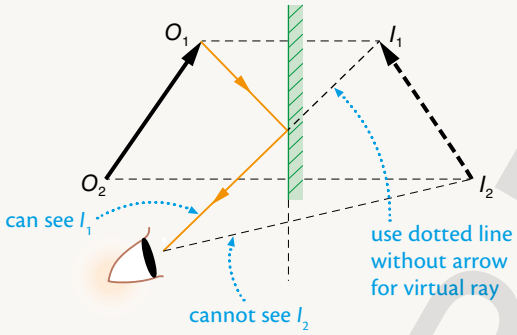
Use these properties to locate the image:



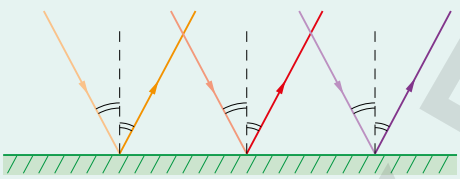
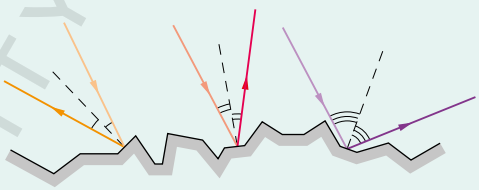


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6 Seeing images



7 Regular reflection and diffuse reflection

Regular reflection	vs	Diffuse reflection
		
smooth	**Surface to occur**	rough
parallel to each other	**Normals**	point in different directions
clear	**Image formed**	not clear
obey	**Laws of reflection**	obey

Key terms

1 angle of incidence 入射角 p.6
2 angle of reflection 反射角 p.6
3 cone of rays 光錐 p.4
4 convergent 會聚 p.3
5 diffuse reflection 漫反射 p.22
6 divergent 發散 p.3
7 image 像 p.10
8 incident ray 入射線 p.6
9 laterally inverted 橫向倒置 p.13
10 laws of reflection 反射定律 p.7
11 normal 法線 p.6
12 parallel 平行 p.3
13 plane mirror 平面鏡 p.6
14 ray 光線 p.3
15 reflected ray 反射線 p.6
16 reflection 反射 p.6
17 regular reflection 單向反射 p.22
18 virtual image 虛像 p.11

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Revision 1

Multiple-choice questions

★1 1.2 A boy walks 2 m away from a mirror. The distance between him and his image

A increases by 2 m.

B increases by 4 m.

C decreases by 2 m.

D decreases by 4 m.

★2 1.2 Mike is 1 m in front of a mirror mounted on a wall. Vicky stands near him. Figure a shows their top view.

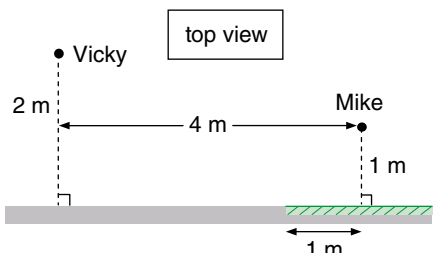


Fig a

Which of the following statements is/are correct?

(1) Mike's image is 5 m from Vicky.

(2) Mike can see that his own image is laterally inverted.

(3) Vicky cannot see Mike's image.

A (2) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

★3 Misc Light is reflected by a smooth, polished metal surface. Which of the following statements about the reflected rays is/are correct?

(1) Parallel rays remain parallel after reflection.

(2) Convergent rays become parallel after reflection.

(3) Divergent rays become convergent after reflection.

A (1) only

B (1) and (2) only

C (1) and (3) only

D (2) and (3) only

★4 1.2 Peter is sitting on the left side of the classroom. He is affected by the glare (the strong light reflected by the whiteboard) and cannot clearly see the words on the whiteboard (Fig b).



Fig b

Which of the following statements about this undesirable situation is/are correct?

(1) The glare is caused by diffuse reflection on the whiteboard.

(2) The glare will be reduced if the windows are covered with curtains.

(3) The glare will be reduced if the board is replaced by another one with a rough surface.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

★5 1.2 Figure c shows the top view of a room 4 m long and 3 m wide. A girl standing at corner *O* is looking at a mirror *PQ* of width 1 m hung on the opposite wall.

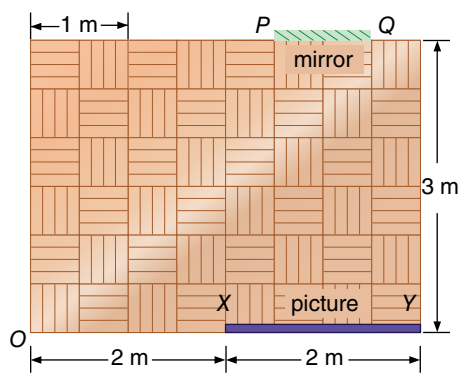


Fig c

What is the minimum width the mirror needs to be extended out from side *P* so that the whole image of *XY* can be seen by the girl?

A 0.5 m B 1 m

C 1.5 m D 2 m

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★6 1.2 A laser beam travels through a set of vertical and horizontal mirrors and reaches the detector (Fig d). Find θ.

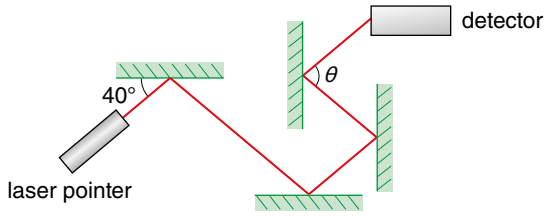


Fig d

A 50°
B 80°
C 90°
D 100°

★7 1.2 An object O is placed at $x = 1$ between two vertical mirrors M_1 and M_2 at $x = -2$ and $x = 2$ respectively (Fig e). I_1 is the image of O formed by M_1. I_2 is the image of O formed by M_2. I_3 is the image of I_2 formed by M_1. Where is I_3?

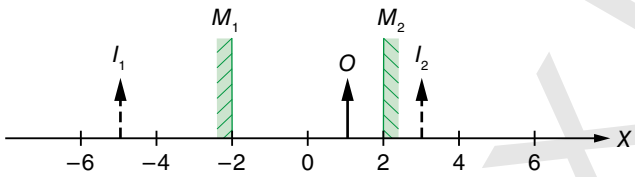
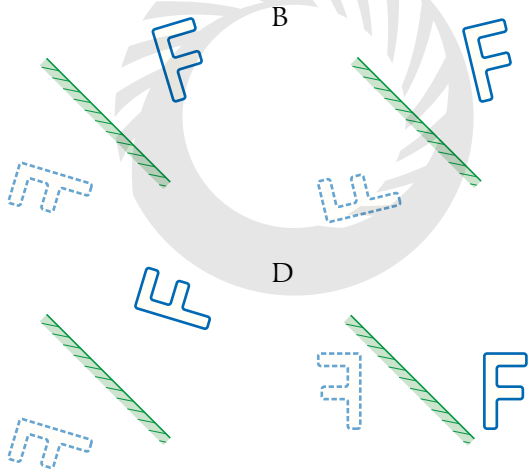


Fig e

A At $x = -3$
B At $x = -5$
C At $x = -7$
D At $x = 5$

★8 1.2 Which of the following diagrams correctly shows the image of F formed by the plane mirror?

A



B

C

D

★9 1.1 Which of the following statements about the image formed by a plane mirror is correct?

A The image becomes smaller when the object moves away from the mirror.
B The image reflects light into our eyes so that we can see it.
C The image emits light into our eyes so that we can see it.
D When we see the image, no light ray comes from the image.

★★10 1.2 Nancy tries to use a small plane mirror to observe a man 2.4 m behind her (Fig g). The man is 1.8 m tall. The mirror is 10 cm long. Nancy's eyes are 1.5 m above the ground. Assume Nancy holds the mirror vertically. What is the maximum distance between Nancy and the mirror for her to see the whole image of the man?

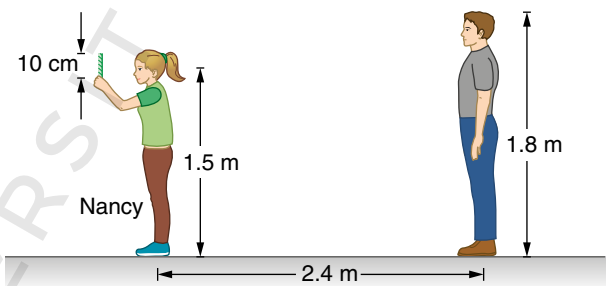


Fig g

A 13.3 cm B 14.1 cm
C 15 cm D 18 cm

▶ Refer p.18, 20

★★11 1.2 A light ray is reflected by a plane mirror. The angle between the incident ray and the reflected ray is 60° (Fig h). After the mirror is turned by an angle θ, the angle between the incident ray and the reflected ray becomes 40°. What is θ? In which direction is the mirror turned?

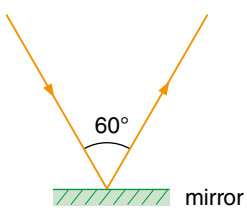


Fig h

A 10° (clockwise) B 20° (clockwise)
C 10° (anticlockwise) D 20° (anticlockwise)

▶ Refer p.8

**12 1.2

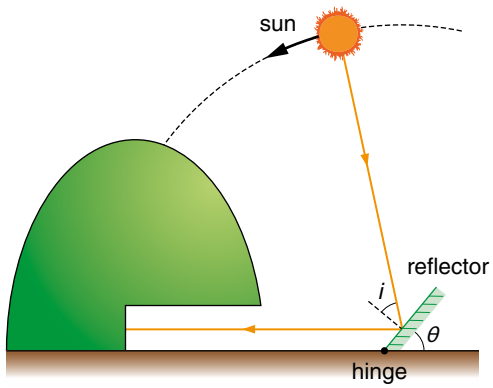


Fig i

Sunlight is incident on a reflector at an angle i to illuminate a cave (Fig h). The reflector can be turned about the hinge and makes an angle θ with the ground. The sun moves along the path in the sky in the direction as shown. To keep the cave illuminated, how will i and θ change?

	i	θ
A	increase	decrease
B	increase	increase
C	decrease	decrease
D	decrease	increase

Refer p.8

13 **HKDSE 2020 Paper 1A Q17** 1.2

Figure i shows a periscope designed by a student. An object is observed via the periscope.

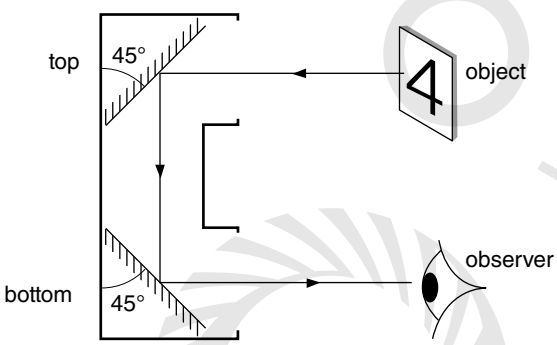
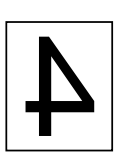


Fig i

Which image will the observer see?

A 

B 

C 

D (41%) 

Conventional questions

*14 1.2 An apple is put on a wooden table (Fig j). No image of the apple is formed by reflection on the table.



Fig j

(a) Name the kind of reflection that occurs on the table. (1 mark)

(b) State the laws of reflection. (2 marks)

(c) (i) Do the laws of reflection apply to the reflection occurring on the table? (1 mark)

(ii) Explain why an image cannot be formed by reflection on the table. (2 marks)

*15 1.2 A ruler is placed in front of a plane mirror and an observer look at the mirror as shown in Figure k.

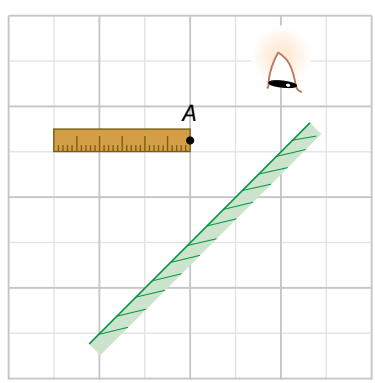


Fig k

(a) Locate and sketch the image of the ruler in Figure k. (1 mark)

(b) Draw a light ray to show how the observer sees the image of point A. Mark A' as the image of A. (3 marks)

(c) If the ruler is moved to the left, in which direction does the image move? (1 mark)

(d) Can the observer see his own image from the position shown? (1 mark)

*16 1.2 A student wants to verify the laws of reflection using the following set-up (Fig l). He directs a light beam towards a plane mirror at different angles of incidence.

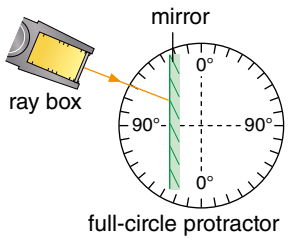


Fig l

(a) State the laws of reflection. (2 marks)

(b) State all the mistakes found in his set-up. (2 marks)

*17 1.2 Figure m shows a rain puddle with a clear image of a building. A pedestrian is passing by.



Fig m

(a) Does regular or diffuse reflection occur on the surface of the puddle? (1 mark)

(b) State the requirement for the phenomenon in (a) to occur. (1 mark)

(c) Figure n shows the situation in Figure m with AB as the surface of the puddle and XY as the pedestrian. Draw the image of the pedestrian and complete the paths of rays r and s. (3 marks)

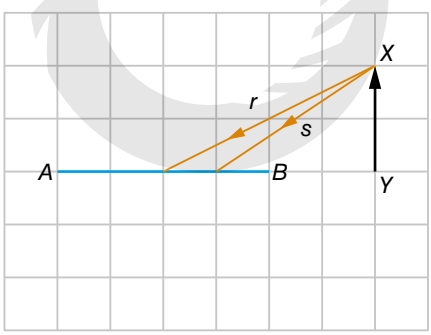


Fig n

*18 1.2 A boy stands in front of a mirror (Fig o).

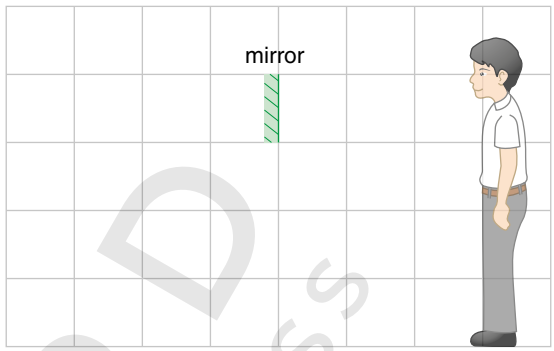


Fig o

(a) Can he see the image of his shoes in the mirror? (1 mark)

(b) Highlight the part of his body that he can see in the mirror. (1 mark)

(c) Can he see the image of his shoes in the mirror if he walks towards the mirror? Draw a diagram to explain your answer. (2 marks)

(d) Suggest two methods that allow him to see the image of his shoes in the mirror. (2 marks)

**19 1.2 Two plane mirrors are facing each other at a right angle. An object O is placed between the mirrors (Fig p). Three images I_1, I_2 and I_3 of O are formed.

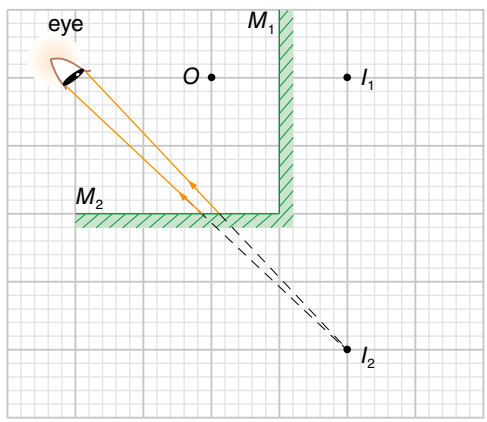


Fig p

(a) Locate I_3 in Figure p. (1 mark)

(b) An eye is looking at I_2. Two rays which appear to come from I_2 enter the eye. By completing the ray diagram, show that the rays come from O after reflection by each mirror in turn. (2 marks)

[Hint: You can regard I_2 as the image of I_1.]

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(c) Show that if an incident ray is reflected by both mirrors, the reflected ray is parallel to the incident ray. (3 marks)

Refer p.16

**20 1.2 A simple periscope consists of a tube and two parallel plane mirrors, M_1 and M_2, each at an angle of 45° to the tube (Fig q).

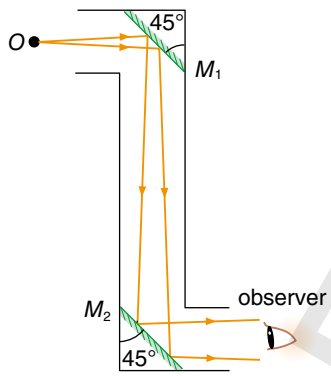


Fig q

(a) In Figure q,

(i) locate the image of O formed by M_1. Mark the image I. (1 mark)

(ii) locate the final image of O formed by M_2. Mark the image I'. (1 mark)

[Hint: You may regard I as an object and I' as the image of I formed by M_2.]

(b) Is the image seen by the observer laterally inverted? Explain. (2 marks)

(c) A similar periscope is installed in a submarine for monitoring the enemy. The distance between the two mirrors is 4 m. Through the periscope, an officer sees a battleship that appears 275 m from his position. What is the actual horizontal distance between the battleship and the officer's position? (1 mark)

Refer p.15, 22

**21 1.2 Betty tries on shoes in a shoe store. She can just see the tip of the shoes at the bottom edge of a plane mirror on the floor (Fig r). The mirror makes an angle of 70° with the floor. Betty's eyes are 1.4 m above the floor.

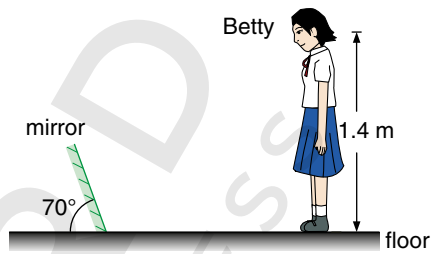


Fig r

(a) Draw a ray in Figure s to show how Betty sees the image of the tip of her shoes. (2 marks)

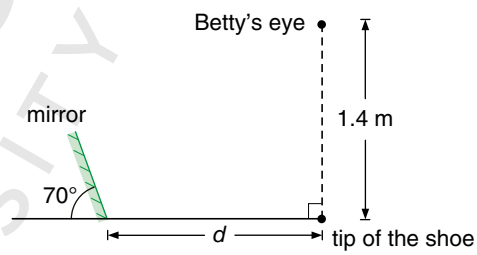


Fig s

(b) Find distance d. (2 marks)

(c) How far is the image of the tip of her shoes from her eye? (2 marks)

(d) The inclination of the mirror is adjusted. After adjustment, Betty cannot see the tip of her shoes if d is greater than 45 cm. What is the angle of the mirror with the floor? (2 marks)

Refer p.18

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Know more

Alhazen (965–1039) (p.2)

In the past, people thought that we could see things because light was emitted by our eyes and bounced off the object. However, in the experiment (similar to the one shown in Fig 1.1a on p.2 but with laterns as the light sources) carried out by Alhazen (Fig a), he found that when an obstacle was placed between one lantern and the hole, the corresponding light spot disappeared. The spot reappeared when the obstacle was removed. This gave evidence that light does not originate from human eye, but rather is emitted from other objects, such as a lantern.



Fig a

Transparent plate (p.12)

When light falls on a transparent plate (e.g. a glass plate), part of it passes through the plate and part of it is reflected (Fig b). Therefore, sometimes when you look at a window, you can see both the objects outside and the images of the objects inside (Fig c). The image is clear when the inside is much brighter than the outside. On the contrary, the image can hardly be seen when the outside is much brighter than the inside.

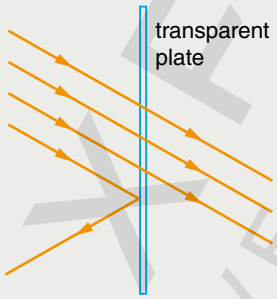
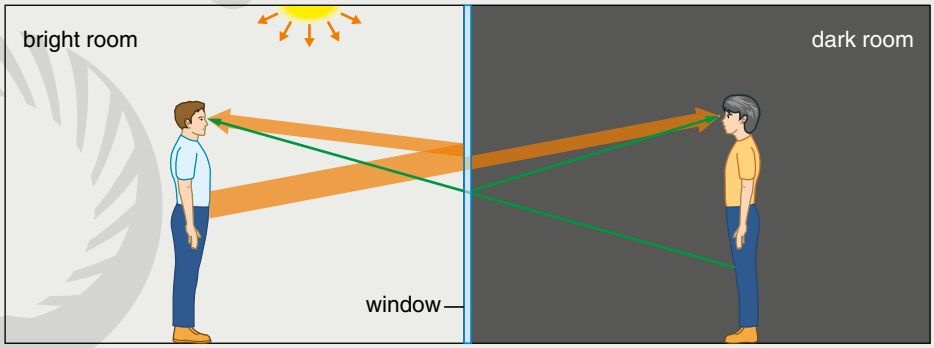


Fig b



Fig c

This principle is used in one-way mirror. It is basically a window installed between a bright and a dark rooms. People in the bright room can only see the images of the objects in this room on the window, while people in the dark room see the objects in the bright room through the window (Fig d).



Light reflected by the window is much stronger than light from the dark room.
⇒ People can only see images of objects in the bright room.

Light from the bright room is much stronger than light reflected by the window.
⇒ People can see objects in the bright room.

Fig d

Know more

Location of an image and laws of reflection (p.12)

An object *O* is placed in front of the mirror *XY*. Consider two light rays from *O* which are reflected by the mirror as shown (Fig a). The reflected rays are perceived by the observer to come from the image *I* behind the mirror.

Consider $\triangle OXY$ and $\triangle IXY$.

① $a = 90° - i$
$= 90° - r$ (law of reflection $r = i$)
$= 90° - c$ (vert. opp. $\angle$s)
$= b$

② $\angle IXY = 180° - \angle OXY$ (adj. $\angle$s on st. line)
$= 180° - 90°$
$= 90°$
$\therefore \angle OXY = \angle IXY$

③ $XY = XY$ (common side)
$\therefore \triangle OXY \equiv \triangle IXY$

Hence $OX = IX$ and $OI \perp XY$.

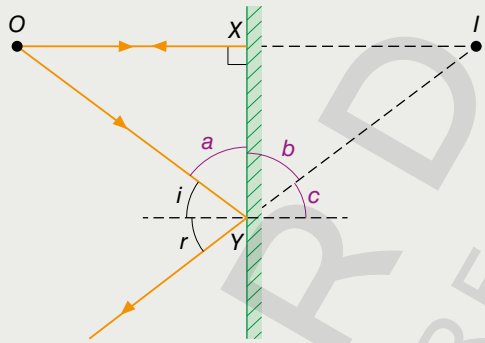


Fig a

Glare (p.24)

Due to the painting or special coating on the surface, the blackboard is quite smooth. When a beam of light is directed onto such surface, although most of the light undergoes diffuse reflection, a small portion of the light undergoes regular reflection (Fig b).

Usually, the light sources in a classroom are located near the ceiling. When light falls on the blackboard, diffuse reflection allows students at different positions to see the words clearly (Fig c).

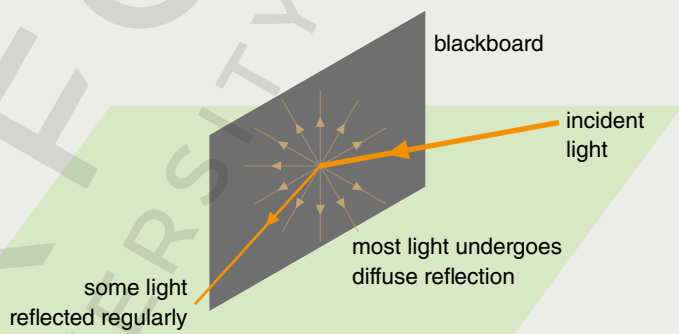


Fig b

Although some light is reflected regularly downwards to the ground, no student is affected because none of them looks at the blackboard from the ground.

However, when there is strong outdoor light through windows, a significant amount of the incoming light is reflected regularly (Fig d). Students at positions along the direction of regular reflection will see glare and cannot read the words on the blackboard.

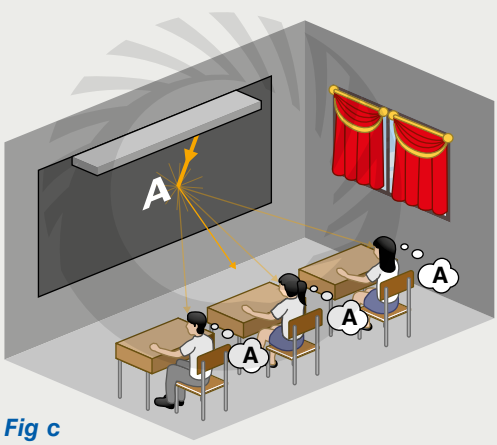


Fig c

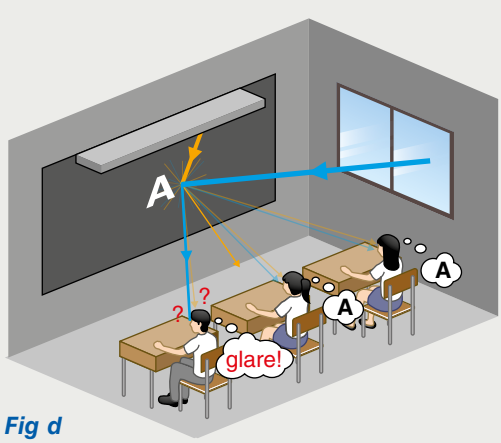


Fig d

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